
G Vibu Vignesh

Systems Engineer (Rust)

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About

Systems Engineer specializing in Rust and low-level Linux kernel primitives. Experienced in building high-performance cloud-native infrastructure, including OCI-compliant container runtimes, CRI protocols, and custom process isolation layers (cgroups v2, namespaces). Combines deep systems programming expertise with a strong background in deploying optimized AI workloads, containerization, and contributing to core open-source OCI tooling.

Experience

Systems Engineer Rust *Tinkerblox Inc | 2025-10 - Present*

Developed oci low level container runtime to deploy applications in linux namespaces that is a drop-in replacement for runc and also worked on cri-protocol grpc that can communicate with kubernetes and deploy applications in cloud infrastructure

- Engineered a high-performance, OCI-compliant container runtime in Rust as a drop-in replacement for runc, significantly accelerating container startup and execution times.
- Architected a gRPC-based Container Runtime Interface (CRI) protocol to enable seamless communication with Kubernetes and orchestrate workloads across cloud infrastructure.
- Leveraged Rust's memory safety to implement low-level Linux primitives (clone, pivot_root, unshare) via nix and libc crates, reducing system call latency without garbage collection overhead.
- Enforced strict process isolation and predictable resource allocation by implementing Linux Namespaces and Cgroups v2.
- Ensured full OCI specification parity to guarantee seamless compatibility and integration with existing container engines and orchestration workflows.

Skills

Systems Programming

Linux Kernel Architecture

Cloud Infrastructure & Kubernetes Integration

GenAI Developer *DoaZ (Remote) | 2024-04 - 2025-10*

Developed and deployed a Retrieval-Augmented Generation (RAG) chatbot to provide rapid answers from complex Korean Construction Rules & Regulation documents.

- Improved information retrieval accuracy from 60% to 80% by implementing Contextual Chunking for document preprocessing.
- Utilized DSPy for advanced prompt optimization, enhancing chatbot response quality and relevance.

- Deployed open-source LLMs (Qwen 148 & 32B) on Vastai GPU servers using VLLM for efficient inference.
- Built and maintained FastAPI endpoints for seamless interaction between the AI backend and the frontend application.
- Deployed LLMs using candle-rust library for faster LLM inference using Rust Programming Language
- Deployed web-applications using Docker to AWS

Skills

Python

RAG Applications

LLM Agents

Rust

Docker

RAG Application Developer *Upwork (Remote) | 2023-11 - 2024-04*

Built a rag chatbot that searched through pdf files and csv files to answer user questions

- Engineered a PDF chatbot featuring a unique user-driven document selection architecture prior to Q&A.
- Developed a CSV-based chatbot leveraging Langchain's Pandas agent framework for data interaction and analysis.
- Created a web-search-enabled chatbot integrating DuckDuckGo to answer queries with real-time information.
- Enhanced overall chatbot functionality using Python and various OpenAI models.

Skills

Python

RAG Applications

Data Scientist Intern *Gilbert Research Institute (Coimbatore) | 2023-08 - 2023-11*

Learnt ML concepts and built Lung and Liver Tumor Detection ML models

- Developed a medical image analysis solution for Lung and Liver Tumor segmentation.
- Utilized PyTorch and Convolutional Neural Networks (CNNs) to build and train the segmentation models.
- Communicated effectively with faculty and staff, incorporating feedback for improvement.
- Collaborated within teams to analyze problems and develop data-driven solutions.

Skills

Python

Deep Learning

Open Source Contributions

[gpt-researcher](#)

[View Pull Request →](#)

Contributed a fix to GPT-Researcher's RAG pipeline, enabling functional metadata filtering by correcting parameter propagation between the research agent and vector store wrappers

Skills

Python

LLM Agents

[youki](#)

[View Pull Request →](#)

Added support for the new `SECCOMP\_FILTER\_FLAG\_WAIT\_KILLABLE\_RECV` flag in youki to align with OCI runtime spec v1.3 updates.

oci-spec-rs

[View Pull Request →](#)

Added `SECCOMP\_FILTER\_FLAG\_WAIT\_KILLABLE\_RECV` to the seccomp configuration to support OCI Runtime Spec v1.3.0 and improve killable wait behavior in seccomp user notifications.

Skills

Systems Programming

Linux Kernel Architecture

Cloud Infrastructure & Kubernetes Integration

Python

RAG Applications

LLM Agents

Rust

Docker

Deep Learning

Education

Integrated B.Sc. & M.Sc.

Indian Institute Of Technology Madras

Languages

English

Tamil

Hindi